

Software Requirements Specification (SRS) Template

Project: Blood Bank Management System

Version: 1.0

Authors: Gurudath A N (PES2UG24CS633); M Sangamesh Prasad (PES2UG24CS640); R Sai Pranav Reddy (PES2UG24CS651); Abhiram Kolalapudi (PES2UG24CS916)

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Revision history

Version	Date	Author	Change summary	Approval
1.0	12-09-2026	Project Team	SRS adapted for Blood Bank Management System	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Blood Bank Management System (BBMS). It defines functional and non-functional requirements, interfaces, security requirements, and verification criteria for managing blood donation, inventory, requests, issue, and monitoring.

1.2 Scope

Covers donor and user registration, authentication, donor eligibility, donation records, blood/component inventory, blood availability search, hospital requests, approval and issue/dispatch, notifications, reporting, audit logging, and administrative maintenance. Excludes laboratory testing procedures and hospital internal systems beyond defined interfaces.

1.3 Audience

Developers, QA Engineers, System Integrators, Blood Bank Staff, Hospital/Authorized Requesters, System Administrators, and Assessment Evaluators.

1.4 Definitions

List of acronyms: BBMS, UI, API, RBAC, OTP, HTTPS, TLS, DB, UUID.

2. Overall description

2.1 Product perspective

The BBMS is a web-based information system used by donors, authorized requesters, blood bank staff, and administrators. It includes a user interface, backend application services, database storage, authentication and authorization services, inventory management, request processing, reporting, and notification support.

2.2 Major product functions (detailed)

- User registration and authentication
- Donor profile and eligibility management
- Blood donation appointment and donation record management
- Blood/component unit and inventory management
- Blood availability search
- Blood request creation, approval, reservation, and issue/dispatch
- Expiry and low-stock alerts
- Reports, audit logs, and administrative maintenance

2.3 User roles and characteristics (expanded)

- Donor: registers, maintains profile, checks eligibility, books donation slots, and views donation history.
- Recipient / Hospital Staff: searches blood availability and submits authorized

blood requests.

- Blood Bank Staff: records donations, manages inventory, verifies requests, and issues blood.
- System Administrator: manages users, roles, configuration, monitoring, and audit information.

2.4 Operating environment

Web browser on desktop or mobile client, application server using Python (Django) and/or JavaScript (Node.js), relational database, secure HTTPS network, and standard server operating environment.

2.5 Constraints

The system shall protect personal and medical-related information, enforce role-based access, maintain consistent inventory records, prevent issue of unavailable units, and use secure communication and approved authentication mechanisms.

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web interface with clear role-based menus, forms, validation messages, search screens, dashboards, and accessible status notifications.

3.2 Hardware interfaces

- Client computer or mobile device
- Optional barcode/QR scanner for blood-unit identification
- Standard keyboard, mouse, touch input, and display
- Server storage for application and database data
- Optional notification gateway hardware where deployed

3.3 Software interfaces

- Database interface for user, donor, donation, inventory, request, and audit data.
- Authentication/authorization service for login, RBAC, and session management.
- Notification service for email/SMS/app alerts.
- Optional hospital or laboratory APIs for authorized interoperability.

3.4 Communications

- HTTPS/TLS shall be used for browser-to-server communication.
- API calls shall use authenticated requests and secure retry handling.
- Notification messages shall be sent through configured secure service providers.

<< Make sure overall there are at least 15 FRs for overall project, 5 NFRs, 2 security objectives and 5 Security requirements>>

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow BBMS-F-###.

4.1 User & Donor Management

Description: Provide registration, authentication, profile, eligibility, appointment, and donation-record functions for donors and authorized users.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-001	The system shall allow donors and authorized users to register with required details and generate a unique user account.	Functional	High	Users / Operations	AC: Valid mandatory data creates one unique account. Test: TC-REG-01	Database required
BBMS-F-002	The system shall authenticate registered users using valid credentials and reject invalid credentials.	Functional	High	Security / Users	AC: Valid login succeeds; invalid login is denied. Test: TC-AUTH-01	HTTPS required
BBMS-F-003	The system shall enforce role-based permissions for Donor, Requester, Blood Bank Staff, and Administrator roles.	Functional	High	Security / Admin	AC: Unauthorized functions are denied. Test: TC-AUTH-02	RBAC dependency

BBM S-F-004	The system shall allow authorized users to view and update their permitted profile information.	Functional	Medium	Users	AC: Valid updates are stored and displayed. Test: TC-PROF-01	Input validation
BBM S-F-005	The system shall allow donors to record eligibility information and book an available donation slot.	Functional	High	Donor / Staff	AC: Eligible donor can book one available slot. Test: TC-DON-01	Eligibility rules required
BBM S-F-006	The system shall record accepted blood donations with donor, date, blood group, component, and unit details.	Functional	High	Blood Bank Staff	AC: Complete donation record creates an identifiable unit. Test: TC-DON-02	Unique unit ID required
BBM S-F-007	The system shall create or update blood/component inventory when a valid donation is accepted.	Functional	High	Operations	AC: Stock count and unit status are updated. Test: TC-INV-01	Transaction consistency
BBM S-F-008	The system shall display current blood availability by blood	Functional	High	Requesters / Staff	AC: Search shows only available	Inventory data required

	group, component, location, and unit status to authorized users.				units matching filters. Test: TC-INV-02	
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4.2 Inventory & Blood Requests

Description: Maintain blood/component stock, provide availability search, process requests, reserve compatible units, and support issue/dispatch.

Req ID	Requirement (shall...)	Type	Priority	Source/STakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-009	The system shall allow authorized hospital/request users to submit blood requests with patient and requirement details.	Functional	High	Hospital / Requester	AC: Complete request receives a unique request ID. Test: TC-REQ-01	Authorization required
BBMS-F-010	The system shall allow blood bank staff to approve or reject a submitted request	Functional	High	Blood Bank Staff	AC: Decision and timestamp are recorded . Test: TC-REQ-02	Staff permission required

	with a recorded decision.					
BBMS-F-011	The system shall reserve compatible available units for an approved request.	Functional	High	Operations	AC: Reserved units are unavailable to other allocations. Test: TC-REQ-03	Compatibility rules required
BBMS-F-012	The system shall record issue/dispatch of reserved blood units and update inventory status and quantity.	Functional	High	Blood Bank Staff	AC: Issued units leave available stock and are traceable to request. Test: TC-ISS-01	Atomic update required
BBMS-F-013	The system shall prevent issue of expired, unavailable, or incompatible blood units.	Functional	High	Safety / Operations	AC: Invalid units cannot be issued. Test: TC-ISS-02	Validation required
BBMS-F-014	The system shall generate alerts for low stock, upcoming expiry, and important request	Functional	Medium	Staff / Admin	AC: Configured alert conditions generate notifications. Test: TC-ALT-01	Notification service

	status changes.					
BBMS-F-015	The system shall provide reports, audit logs, and administrative user/role management to authorized administrators.	Functional	Medium	Administrator	AC: Authorized admin can generate report and review audit trail. Test: TC-ADM-01	Audit storage required

4.3 Donation Management

Requirements: The system shall record donation details, associate collected units with donors, capture blood group/component information, and update inventory after accepted donations. IDs: BBMS-F-005..BBMS-F-007.

4.4 Blood Issue / Dispatch

Requirements: Support authorized blood requests, approval/rejection, compatibility checks, reservation, issue/dispatch, stock update, and confirmation. IDs BBMS-F-010..BBMS-F-013.

4.5 Maintenance & Reporting

Requirements: Provide alerts, reports, audit logs, user/role administration, and system monitoring for authorized staff and administrators. IDs BBMS-F-014..BBMS-F-015.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs BBMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
BBMS-NF-001	95% of normal user page/API requests shall complete within 3	Performance	High	95th percentile <= 3s in performance test. Test: TC-PERF-01

	seconds under representative load.			
BBMS-NF-002	The system shall provide at least 99.5% monthly availability excluding approved maintenance windows.	Reliability	High	Monthly uptime >= 99.5%. Test: TC-REL-01
BBMS-NF-003	The system shall preserve data integrity so that user, inventory, request, and issue transactions remain consistent.	Data Integrity	High	Concurrency and rollback tests show no inconsistent stock or duplicate issue. Test: TC-INT-01
BBMS-NF-004	The application shall support at least 200 concurrent active users without critical functional degradation.	Scalability	Medium	Load test meets concurrency target. Test: TC-SCAL-01
BBMS-NF-005	The interface shall provide clear validation, accessible navigation, and responsive screens for	Usability / Accessibility	Medium	Usability review and accessibility checklist pass. Test: TC-UX-01

	supported devices.			
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5.1. Security

5.1.1 Security Objectives

The security objectives are to protect confidential user and blood-bank data, preserve the integrity and traceability of inventory and requests, and ensure that only authorized roles can perform sensitive operations.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
BBMS-SR-001	All browser-to-server and API communications shall use HTTPS/TLS.	Security	High	Network scan confirms encrypted transport. TC-SEC-01
BBMS-SR-002	Passwords shall be stored only using a secure one-way password hashing mechanism and shall never be stored in plaintext.	Security	High	Database inspection confirms no plaintext passwords. TC-SEC-02
BBMS-SR-003	Role-based access control shall restrict donor, requester, staff, and administrator operations according to assigned permissions.	Security	High	Unauthorized access attempts are denied and logged. TC-SEC-03
BBMS-SR-004	The system shall enforce session timeout, logout, and secure session invalidation after configured inactivity.	Security	Medium	Idle session becomes invalid and protected pages require re-authentication. TC-SEC-04
BBMS-SR-005	Audit logs shall record critical user, inventory, approval, issue, and administrative actions with actor and timestamp.	Security	High	Critical actions are traceable in audit logs. TC-SEC-05

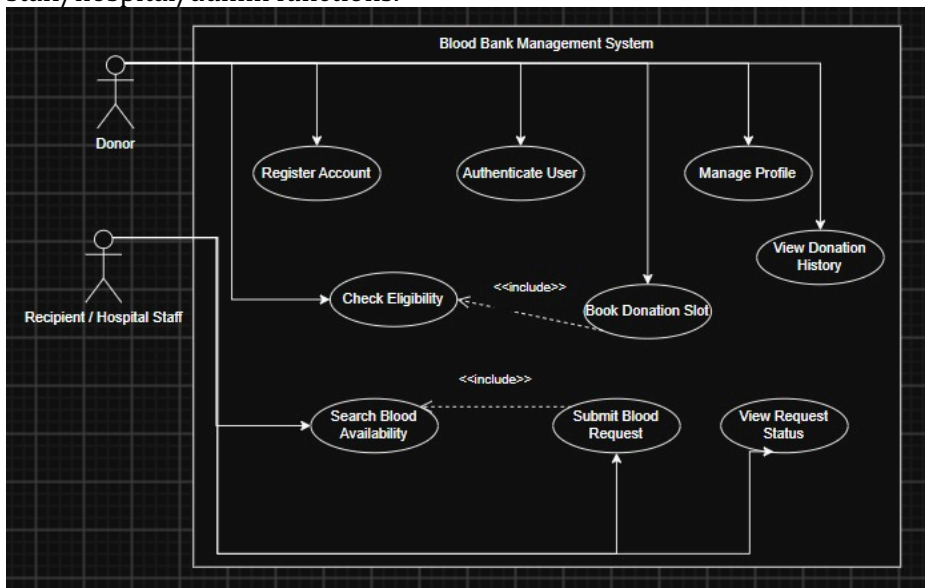
6. Quality attributes & Acceptance tests

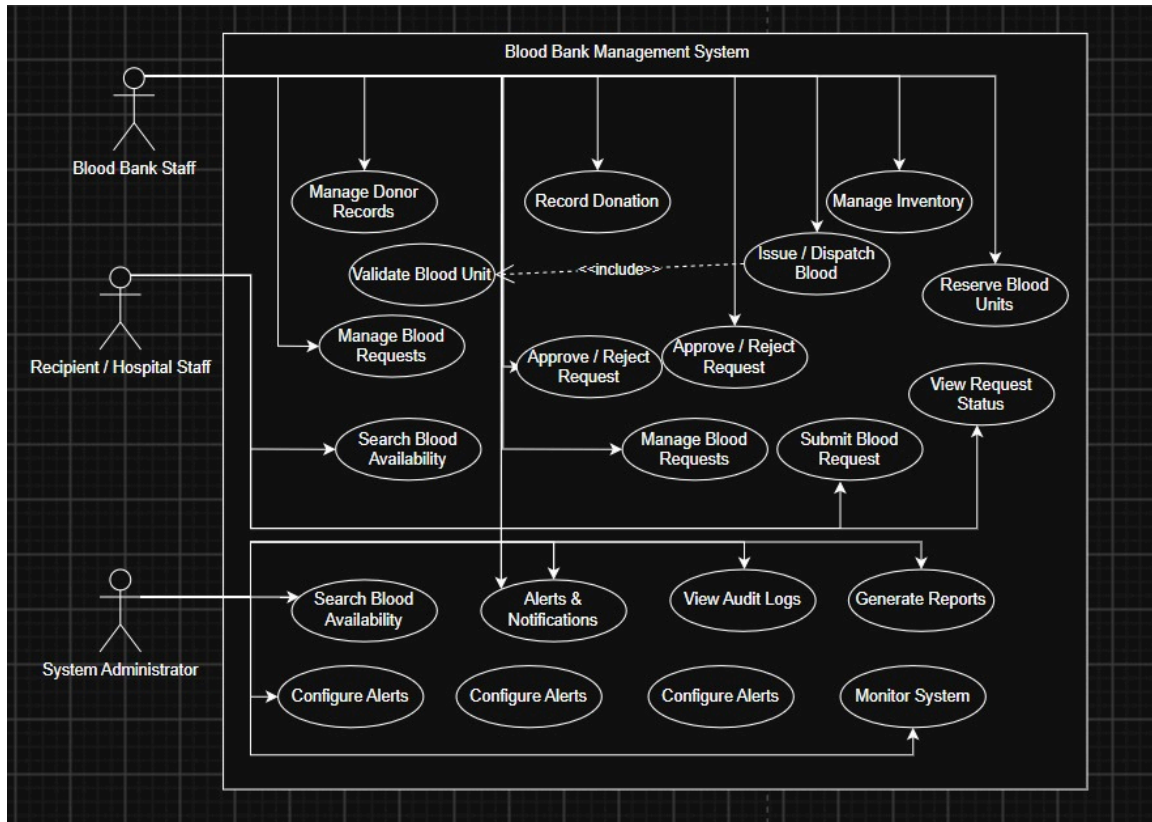
- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR or security failures, and RTM shows all required test cases passed.
- Acceptance test suites: Registration & Authentication, Donor & Donation, Inventory, Blood Request & Issue, Performance, Security, Accessibility, and Audit tests.

7. System models and diagrams

7.1 UML Use-Case diagram

At least 2 UML use case diagrams are included for donor/recipient functions and staff/hospital/admin functions.





8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
BBMS-F-001	allow donors and authorized	4 / DS-AuthModule	AuthModule	TC-REG-01	N	
BBMS-F-002	authenticate registered user	4 / DS-AuthModule	AuthModule	TC-AUTH-01	N	
BBMS-F-003	enforce role-based permission	4 / DS-RBACModule	RBACModule	TC-AUTH-02	N	
BBMS-F-004	allow authorize	4 / DS-ProfileModule	ProfileModule	TC-PROF-01	N	

	d users to vi					
BBMS-F-005	allow donors to record eligi	4 / DS- Donation Module	Donation Module	TC- DON-01	N	
BBMS-F-006	record accepted blood donati	4 / DS- Donation Module	Donation Module	TC- DON-02	N	
BBMS-F-007	create or update blood/co mpo	4 / DS- Inventory Module	Inventory Module	TC-INV- 01	N	
BBMS-F-008	display current blood availa	4 / DS- Inventory Module	Inventory Module	TC-INV- 02	N	
BBMS-F-009	allow authorize d hospital/r e	4 / DS- Request Module	Request Module	TC-REQ- 01	N	
BBMS-F-010	allow blood bank staff to ap	4 / DS- Request Module	Request Module	TC-REQ- 02	N	
BBMS-F-011	reserve compatib le available	4 / DS- Allocatio nModule	Allocatio nModule	TC-REQ- 03	N	
BBMS-F-012	record issue/dis patch of res	4 / DS- IssueMo dule	IssueMo dule	TC-ISS- 01	N	
BBMS-F-013	prevent issue of expired, un	4 / DS- IssueMo dule	IssueMo dule	TC-ISS- 02	N	
BBMS-F-014	generate alerts for low stoc	4 / DS- AlertMod ule	AlertMod ule	TC-ALT- 01	N	
BBMS-F-015	provide reports,	4 / DS- AdminM odule	AdminM odule	TC- ADM-01	N	

	audit logs,					
BBMS-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-PERF-01	N	
BBMS-NF-002	Availability target	5 / DS-Rel-01	Infrastructure	TC-REL-01	N	
BBMS-NF-003	Data integrity	5 / DS-Int-01	DB / API	TC-INT-01	N	
BBMS-NF-004	Concurrency target	5 / DS-Scal-01	WebUI / API	TC-SCAL-01	N	
BBMS-NF-005	Usability & accessibility	5 / DS-UX-01	WebUI	TC-UX-01	N	
BBMS-SR-001	Secure transport	5.1.2 / DS-Sec-01	Security Module	TC-SEC-01	N	
BBMS-SR-002	Password hashing	5.1.2 / DS-Sec-02	AuthModule	TC-SEC-02	N	
BBMS-SR-003	RBAC enforcement	5.1.2 / DS-Sec-03	RBACModule	TC-SEC-03	N	
BBMS-SR-004	Session security	5.1.2 / DS-Sec-04	AuthModule	TC-SEC-04	N	
BBMS-SR-005	Audit logging	5.1.2 / DS-Sec-05	AuditModule	TC-SEC-05	N	